

JavaScript Basics Assessment — Answer Key

Question 1 — let · var · const

Part A

Which statement about const is correct?

- A. A const variable can be reassigned at any time.
- B. const variables are function-scoped.
- **C. A const variable must be assigned a value when declared.**
- D. const prevents any change to an object's properties.

Answer: C

Part B

Fill in the blanks with let, var, or const:

- Use _____ when the value will never change.
- Use _____ when the value needs to change inside a block.

Answer i: const

Answer ii: let

Part C

What will this code print?

```
var x = 10;
if (true) {
  var x = 20;
  let y = 30;
}
console.log(x);
console.log(y);
```

Answer:

20

ReferenceError: y is not defined

Explanation: var x is function-scoped (or globally scoped here), so reassigning it inside the if-block changes the value of x to 20. However, let y is block-scoped to the if-statement, meaning it does not exist outside of it, causing a ReferenceError when console.log(y) is called.

Question 2 — if · else if · nested if-else

Part A

What is printed when score = 55?

```
if (score >= 90) { console.log("A"); }  
else if (score >= 75) { console.log("B"); }  
else if (score >= 50) { console.log("C"); }  
else { console.log("F"); }
```

- A. A
- B. B
- **C. C**
- D. F

Answer: C

Part B

Write a nested if-else for these age categories:

```
age < 13 → print "child"  
age < 18 → print "teenager"  
age < 60 → print "adult"  
otherwise → print "senior"
```

Answer:

```
if (age < 13) {  
    console.log("child");  
} else {  
    if (age < 18) {  
        console.log("teenager");  
    } else {  
        if (age < 60) {  
            console.log("adult");  
        } else {  
            console.log("senior");  
        }  
    }  
}
```

(Note: A standard chaining syntax using else if is also conceptually correct, but if strict nesting is required, the above structure satisfies it.)

Question 3 — Ternary operator

Part A

What is the value of result?

```
let num = 8;  
let result = (num % 2 === 0) ? "even" : "odd";  
console.log(result);
```

Answer: "even"

Part B

Rewrite this as a single ternary expression stored in a variable called access:

```
if (loggedIn) { access = "welcome!"; }  
else { access = "please log in"; }
```

Answer:

```
let access = loggedIn ? "welcome!" : "please log in";
```

Question 4 — for · while · do-while

Part A

Which loop always runs at least once, even if the condition is false?

- A. for loop
- B. while loop
- **C. do-while loop**
- D. None of the above

Answer: C

Part B

What does this loop print?

```
let i = 1;  
while (i <= 5) {  
  if (i % 2 !== 0) { console.log(i); }  
  i++;  
}
```

Answer:

1
3
5

Part C

Write a for loop that prints numbers from 10 down to 1.

Answer:

```
for (let i = 10; i >= 1; i--) {  
  console.log(i);  
}
```